OFX Legends Re-Slot Analysis

TO-BE Report

Submitted to:

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Executive Summary

The Publix Pharmacy Distribution Center (OFX) is a critical hub supplying mediations to over 1,400 pharmacy locations. Within this building, the Legends selection area handles high-volume manual picking across 3,021 items stored in 5,040 rack locations. The current slotting configuration for the items has not received a major revision since the facility opened in 2016, resulting in measurable inefficiencies in labor utilization, congestion, and ergonomics.

Analysis of a full-year dataset showed that item placement is misaligned with operational priorities. High demand products are frequently located in ergonomically unfavorable positions, while lower demand items occupy prime picking zones. This misalignment increases pick time, contributes to aisle congestion, and exposes associates to unnecessary physical strain.

The current system also exhibits poor workload distribution across aisles, leading to localized congestion and underutilized capacity elsewhere. Container sequencing issues further introduce product damage risk, particularly where boxes are improperly positioned relative to bottles. Collectively, these inefficiencies stem from the absence of a data-driven slotting methodology that can be continuously updated as needs change.

Despite these challenges, the facility presents a strong opportunity for improvement. With approximately 40% of rack space unused, the system can be significantly optimized without requiring capital investment or structural changes. The analysis confirms that a redesigned slotting approach, based on item velocity, ergonomics, weight, and congestion balancing, can improve performance across all key metrics.

Recommended improvements include both a targeted re-slotting of high impact items for immediate gains and a comprehensive system-wide slotting method to later achieve maximum efficiency. The solution is designed to reduce total pick time, improve worker safety, and balance workload across aisles, while remaining adaptable for future demand changes.

Implementation of the proposed re-slotting strategy results in measurable performance improvements across all key metrics. Total daily pick time is reduced from 81.77 hours to 76.19 hours, representing a 6.82% improvement. The proportion of picks occurring within the ergonomic golden zone increases from 77.3% to over 90%, while picks from the most physically demanding levels are reduced by approximately 66%. These improvements are achieved by relocating approximately 16% of items, resulting in an estimated annual labor savings of 2,031 hours, equivalent to approximately one FTE. The projected financial impact yields a payback period of approximately 1.3 years and a five-year return on investment exceeding 250%.

From a business perspective, this project represents a low-cost, high-impact opportunity to improve labor productivity, reduce ergonomic risk, and enhance operational efficiency without requiring capital investment. The proposed methodology not only delivers immediate performance gains but also establishes a scalable, data-driven framework for continuous improvement. By implementing this approach, Publix can sustain long-term efficiency improvements, better utilize existing capacity, and maintain high service levels across its pharmacy distribution network.

Introduction

The Publix Pharmacy Distribution Center (OFX), located in Orlando supplies medications to over 1,400 Publix Pharmacy locations across multiple states, supporting both retail and Central Pharmacy operations. Within the facility, the Legends selection area handles thousands of products stored across many rack locations, while associates manually pick to fulfill orders. Due to the high volume of daily activity, how items are arranged directly impacts speed, worker movement, congestion in aisles, and overall labor efficiency. Even small inefficiencies in item placement can add up significantly in a workday.

However, the current slotting layout has not been updated since 2016, despite changes in demand and operational need. The outdated layout leads to unnecessary travel, congestion, and poor labor usage. Improving the slotting plan involves balancing multiple constraints such as weight limit, ergonomic placement, and the movement of the item codes. By using a structured, data-driven approach to reorganize items, the facility can reduce congestion, improve picking efficiency, and better support the stores that it services.

The design of the project is shaped by a set of clear realistic constraints that define what solutions are feasible. Rather than optimizing purely for speed, the slotting planogram must operate within boundaries such as safety regulations, ergonomic limitations, facility layout, and operational workflows. These constraints guide decision making by forcing tradeoffs. As a result, the project becomes more involved with balancing performance improvements with practical limitations, rather than a purely optimization-based problem.

Safety, health, and ethical considerations play a central role in narrowing the scope. Ergonomic slotting, adherence to OSHA standards, and reduced congestion directly protect workers from injury while simultaneously improving productivity. At the same time, the system supports public health by ensuring medications are distributed quickly and reliably. Ethical responsibility is reflected in the use of transparent, data-driven methods, while social and cultural factors ensure that solutions remain intuitive, inclusive, and easy for a diverse workforce to adopt without extensive retraining.

Economic, environmental, and global considerations further guide the solution approach. By leveraging existing infrastructure and avoiding costly automation, the design remains financially feasible while still delivering measurable labor savings. Reduced travel distances and improved efficiency lower energy usage, supporting environmental sustainability without requiring major capital investment. Additionally, the methodology is scalable and transferable, meaning the same principles can be applied across other facilities with minimal adjustment. Together, these constraints ensure the final design is not just optimized for one location, but robust, sustainable, and adaptable across the broader supply chain.

DMAIC: Define

Process Being Examined and Problem Addressed

The process under examination is the manual picking operation within the Legends selection area of the Publix Pharmacy Distribution Center (OFX), located in Orlando, Florida. The facility supplies medications to over 1,400 Publix Pharmacy locations across multiple states. Within the Legends area, approximately 20 selectors operate simultaneously, traversing seven aisles (LA, LB, LC, LD, LE, LF, and LM) to fulfill pharmacy orders from a carton flow rack system. The system is comprised of 5,040 static rack locations distributed across five vertical levels per bay.

The rack structure defines the ergonomic character of every pick. Levels 1 and 2 require floor-level bending and represent the most physically demanding picking positions. Levels 3, 4, and 5, collectively referred to as the golden zone, correspond to mid- to upper-torso reach heights and represent the range of motion most compatible with efficient, low-strain picking. Because selectors interact with these positions hundreds of times per shift, the vertical assignment of items directly governs labor time and injury exposure.

The current slotting configuration has not been substantively revised since the facility opened in 2016. Despite significant changes in item demand over the intervening decade, item codes have not been systematically repositioned to reflect updated velocity profiles or weight characteristics. The consequence is a slotting arrangement that is misaligned with operational priorities across multiple documented dimensions, which are elaborated in the problem statement below.

Problem Statement

The carton flow rack slotting planogram at the Publix OFX Legends selection area has not been updated since 2016, resulting in a systematic misalignment between item placement and item movement velocity. Analysis of a full-year warehouse management system (WMS) dataset revealed that, under the current arrangement, 22.7% of all pick volume occurs at Levels 1 and 2, the most physically demanding positions, while 77.3% originates from the golden zone (Levels 3, 4, and 5). This distribution indicates that a significant portion of daily selector activity requires repeated bending and awkward posture that is avoidable through targeted item repositioning. These conditions generate unnecessary picker strain, increase musculoskeletal injury exposure, and produce suboptimal labor utilization across the facility.

Goal Statement and Project Objectives

The primary goal of this project is to reduce total active picking time and improve ergonomic conditions at the Publix OFX Legends selection area by redesigning the carton flow rack slotting planogram based on item movement data. The improvement is to be achieved through a targeted repositioning of high-movement items from strenuous Levels 1 and 2 into available golden zone positions currently occupied by lower-movement items, using the existing rack infrastructure and without capital investment or structural modification.

The specific project objectives are as follows. First, increase the proportion of total pick volume originating from the golden zone (Levels 3, 4, and 5) from 77.3% to greater than 90%, excluding the LM aisle. Second, reduce the proportion of pick volume occurring at strenuous Levels 1 and 2. Third, reduce overall daily picking time by improving the ergonomic accessibility of the highest-movement items. Fourth, minimize the number of items physically moved during implementation to limit operational disruption while still achieving most of the available improvement. Fifth, present a business case demonstrating that the proposed planogram warrants full-scale internal development by the Publix Industrial Engineering team.

Key Measures to Evaluate Success

The success of the proposed re-slotting intervention will be evaluated against the following key performance metrics, each of which is quantifiable from the project dataset and directly traceable to the problem dimensions identified above.

Key Measure	Baseline (AS-IS)	Target (TO-BE)
Total Pick Time (hrs/day)	81.77	76.19
% of Picks within Golden Zone	77.3%	92%
% of Picks out of Golden Zone	22.7%	8%
Workload Reduction	-	1 FTE
% of Items Moved	-	16% of Total Items

Table 1: Key Performance Measures and Target Values

Scope Statement

This project is scoped to the Legends Carton Flow Rack Selection area within the Publix OFX Pharmacy Distribution Center. The scope encompasses the slotting arrangement of all 3,021 currently occupied rack locations across aisles LA, LB, LC, LD, LE, LF, and LM, at all five rack levels. The project does not address the physical rack infrastructure, the WMS location assignment system, entry and exit points of the picking area, or any other area of the distribution center outside the Legends selection zone.

The solution presented is scoped as a targeted, high impact re-slot rather than a complete planogram overhaul. The approach was deliberately chosen to minimize operational disruption and to remain feasible given the absence of data on whether items currently occupy multiple facing lanes. By repositioning approximately 16% of items, the proposed plan achieves an estimated 65% of the theoretically optimal solution, defined as all non-LM items migrating to the golden zone. This scoped approach allows Publix to pilot the methodology and validate results before committing to a broader system-wide implementation.

Needs, Requirements, and Constraints

The slotting redesign must operate within the following constraints. The solution must comply with the mezzanine structural load limit of 14 lbs per square foot per carton flow level. The bottle-before-box container sequencing rule is a non-negotiable operational requirement that must be satisfied throughout the picking sequence. All relocated items must map to existing rack locations without requiring physical expansion of the rack footprint. The project must be implementable by the existing DC workforce using standard material handling procedures, without specialized equipment or third-party contractors. Finally, the re-slotting plan must be documented at the individual item-code level so that the Publix IE team can execute, verify, and update it as demand patterns change over time.

DMAIC: Measure

The Measure phase establishes a quantitative baseline of the existing slotting arrangement at the Publix OFX Legends selection area. All analyses described in this section are derived from the dataset provided by the Publix Industrial Engineering team, exported on January 23, 2026, covering the preceding 52 calendar weeks of operational activity. The dataset was loaded into the project EDA workbook and subjected to a comprehensive exploratory data analysis to characterize item velocity, rack level pick volume distribution, aisle workload balance, container sequencing, weight placement, and item activity profiles.

Data Collection and Dataset Overview

The primary dataset is a complete inventory snapshot of the carton flow rack selection area, extracted from the DC's warehouse management system as a single cross-sectional query. Each of the 3,021 records represents one item code occupying one rack location and carries attributes including aisle, bay, rack level, average bottles per week, average cases per week, maximum bottles per week, item weight (lbs.), container type (bottle or box), item dimensions (length, width, height in inches), and weeks shipped within the 52-week observation window. Because the data originates from the DC's WMS, measurement error in location codes, dimensions, and weight fields is assumed negligible. Seventy-seven records (2.5% of total) were identified as having no shipping transactions during the observation period and were excluded from rate-based analyses while being retained in occupancy counts.

Parameter	Value	Notes
Total Flowrack Locations	5,040	Fixed physical capacity
Occupied Locations	3,021	59.9% Utilization
Unique Item Codes	3,021	One item code per location
Rack Levels Per Bay	5	1 = Bottom, 5 = Top
Selection Aisles	7	LA, LB, LC, LD, LE, LF, LM
Observation Window	52	Full year
Data Run Date	Jan. 23rd, 2026	Snapshot date
Records Missing Movement Data	77	2.5% of data

Table 2: Data Structure and Key Parameters

Key Performance Variables

Key Performance Output Variables (KPOVs)

The primary KPOV is pick ergonomic efficiency, operationalized as the proportion of total pick volume originating from the golden zone (Levels 3, 4, and 5). A well-slotted system concentrates high-movement items in these levels, minimizing selector bending at Levels 1 and 2. Secondary KPOVs include aisle congestion balance, measured as the distribution of total pick volume across aisles and bays; weight-level alignment, representing the correspondence between item weight and rack level; and location utilization, the ratio of occupied to total available rack locations.

Key Performance Input Variables (KPIVs)

The KPIVs driving slotting decisions are average bottles per week, average cases per week, and minimum replenishment trips required per week. These three metrics were used to calculate a composite percentile rank score for each item code, which served as the primary basis for identifying relocation candidates. The percentile score for each of the three metrics was calculated independently, and the three scores were then averaged to produce a single overall movement rank per item. Items in the top 60th percentile of movement currently assigned to Levels 1 or 2 were identified as candidates for relocation to the golden zone. Items in the bottom 30th percentile of movement currently occupying golden zone positions (Levels 3, 4, and 5) were identified as swap targets. Additional KPIVs considered include item weight, container type, item physical dimensions, and weeks shipped in the observation window.

Current Performance Level

Table 3 presents descriptive statistics for the eight continuous variables in the dataset. The throughput distribution is the most diagnostically important feature. The mean average number of bottles per week (307.6) exceeds the median (72.9) by a factor of more than four, and the standard deviation (894.8) is nearly three times the mean, confirming that the distribution is heavily right-skewed. This extreme skew is operationally significant because it indicates that a small number of high-movement item codes drive a disproportionate share of aggregate pick volume, and that the ergonomic and congestion consequences of their placement are correspondingly outsized. Item weight follows a similar right-skewed profile, with a mean of 7.86 lbs and a maximum of 134.2 lbs. The weeks-shipped variable shows a median of 49 weeks, indicating that most items ship consistently throughout the year, though 104 items shipped in five or fewer weeks, representing near-dormant inventory occupying active pick positions.

Variable	Mean	Median	St. Dev.	Min	Max
Avg. Bottles/Week	307.6	72.9	894.8	0.0	18,152.0
Avg. Cases/Week	11.0	2.3	42.5	0.0	1,771.0
Max Bottles/Week	479.1	117.0	1,461.8	0.0	29,059.0

Item Weight (lbs)	7.9	5.4	7.7	0.0	134.2
Weeks Shipped	42.2	49.0	14.1	1.0	52.0
Length (in)	13.0	12.0	7.4	1.0	155.0
Width (in)	9.6	9.0	6.0	0.5	145.0
Height (in)	7.4	6.2	5.6	0.1	180.0

Table 3: Descriptive Statistics for KPIs

Rack Level Analysis

Table 4 presents the distribution of pick volume by rack level under the current slotting arrangement. This analysis uses the percentage of total bottles picked per level, as captured in the project EDA workbook, as the primary measure of ergonomic alignment. The data confirms that the existing planogram does not enforce alignment between item movement and rack position.

Under the current configuration, 29.7% of all pick volume originates from Level 5, the highest share of any single level. Levels 3, 4, and 5 combined account for 77.3% of total pick volume, meaning that 22.7% of picks occur at Levels 1 and 2, where selectors must repeatedly bend to floor height. Level 2 alone accounts for 22.0% of all pick volume, driven by the presence of high-movement items misassigned to that strenuous position. This distribution provides the direct basis for the improvement methodology: high-movement items at Levels 1 and 2 are identified as relocation candidates, and low-movement items occupying Levels 3, 4, and 5 are identified as swap targets.

Level	Description	Items	% Picks	Fast Items	% Items Fast	Slow Items	% Items Slow
5	Golden Zone	609	29.7%	21	3.4%	277	45.5%
4	Golden Zone	729	23.4%	14	1.9%	343	47.1%
3	Golden Zone	718	24.2%	22	3.1%	328	45.7%
2	Low	727	22.0%	16	2.2%	347	47.7%
1	Floor	219	0.6%	0	0.0%	205	93.6%
Total		**3,002**	**100%**	**73**		**1,500**	

Table 4: Pick Volume Distribution (excluding LM)

Aisle Congestion Analysis

Table 5 summarizes pick volume and utilization by aisle. Congestion risk is assessed by comparing each aisle's total throughput relative to its physical capacity. Aisle LF carries the highest total throughput at 145,486 average bottles per week across 587 items, while Aisle LB carries the lowest at 114,788 bottles per week across 463 items. The 26.8% differential in total pick volume between LF and LB, combined with LF's higher item count, indicates that LF concentrates more selector activity than any other standard aisle while LB retains available capacity to absorb additional high-movement items. The LM aisle is a compact, high-density zone with only 22 total locations, 19 of which are occupied. Its average throughput of bottles per week far exceeds the standard aisles, and it is treated separately from the primary re-slotting analysis given its distinct operational characteristics. Aisle LF represents the highest congestion risk in the system under the current arrangement.

Aisle	Items	Utilization	Avg. Bottles/Week	Congestion Risk

LA	451	80%	129,701.1	
LB	463	72%	114,787.7	Lowest
LC	468	49%	115,705.4	
LD	490	84%	136,324.3	
LE	543	93%	126,988.7	
LF	587	98%	145,485.9	Highest
LM	19	100%	136,493.8	None (Pallet Select)

Table 5: Throughput by Aisle

Item Activity Distribution

Table 6 stratifies the 2,944 records with complete movement data by weeks shipped over the 52-week observation window. The activity profile shows that 45.4% of slotted items (1,336) shipped in all 52 weeks of the observation period and represent the core of stable, year-round demand. These consistently active items are the highest-priority candidates for golden zone placement, as they generate pick volume every week and benefit most from ergonomic positioning. At the opposite extreme, 104 items (3.5%) shipped in five or fewer weeks, functioning as near-dormant inventory that occupies active pick locations without generating meaningful order volume. The 40.1% rack vacancy (2,019 empty locations) confirms that the project team can execute a materially improved slotting plan entirely within the existing physical footprint without requiring any structural changes.

Activity Tier	Weeks Shipped	Item Count	% of Total	Operational Interpretation
Very Low	1-5	104	3.5%	Near Dormant, remove from active picking area.
Low	6-25	333	11.3%	Seasonal demand, active during only part of the year.
Moderate	26-51	1,171	39.8%	Regular, but not year-round demand.
Consistent	52	1,336	45.4%	Year-round demand. Highest priority for golden zone.
Total		**2,944**	**100%**	

Table 6: Item Activity by Weeks Shipped

Target Performance Levels

Based on the current state analysis and the heatmap dashboard and level balance measures developed in the project EDA workbook, the following target performance levels are established for the Improve phase. Total daily pick time is targeted to decrease by approximately 7%, from 81.77 hours per day to 76.19 hours per day, achieved by repositioning approximately 16% of slotted item codes. The proportion of total pick volume originating from the golden zone (Levels 3, 4, and 5) is targeted to increase from 77.3% to greater than 90%, excluding the LM aisle, representing a 20% improvement in ergonomic pick distribution. The proportion of pick volume occurring at strenuous Levels 1 and 2 is targeted to decrease from 22.7% to approximately 8%, a reduction of 66%. The overall workload reduction achieved by the proposed re-slot is estimated to be equivalent to approximately 1 FTE, used here as a scale of measurement rather than a staffing recommendation. Near-dormant items shipping in five or fewer weeks are to be evaluated and removed or relocated from active pick positions to free those locations for higher-frequency item codes.

DMAIC: Analyze

The Analyze phase examines the relationships between the KPIVs and the KPOVs that define slotting performance. The objective is to move from describing the current state to explaining where the root causes of inefficiency are located. The analyses in this section draw directly on the quantitative findings from the Measure phase and apply a structured cause-and-effect framework to connect observable data patterns to actionable slotting deficiencies.

Analytical Approach

The team employed a combination of stratification analysis, cross-tabulation, distribution comparison, and correlation analysis to identify and quantify the root causes of poor slotting performance. Because the dataset is a complete census of the slotted inventory rather than a sample, statistical inference tests are not required to establish the existence of patterns. The task is instead to measure the size and significance of those patterns and map them to actionable changes.

The guiding methodology for each sub-analysis is to minimize picker labor, reduce aisle congestion, prevent packaging damage, and comply with structural constraints. The team compares each analysis to these primary goals of the project and identifies how significant they are to producing the desired results.

Root Cause Analysis

Item Movement Proxy

A key design assumption for the proposed slotting plan is deciding on which of the three movement metrics should be used to determine item slotting priority. Rather than selecting one metric and using that alone, the percentile each item fell into for each of the metrics was calculated, and their average percentile score was used for ranking.

Velocity-Level Mismatch

The most consequential root cause identified in this analysis is the absence of a velocity-driven slotting rule in the current arrangement. In an optimally slotted warehouse, the Pearson correlation between rack level ergonomic cost (highest at levels 1 and 2, lowest at levels 3, 4, and 5) and item velocity would be strongly negative: higher velocity should predict lower ergonomic cost of picking. The current data show no such relationship.

Level 1, the most ergonomically costly position for the picker (requiring lifting from below the knees), holds 14 Fast or Medium-tier items. Each of these items present an opportunity to provide substantial pick time improvements for the system at large. In Level 1, a large majority of item movement occurs from just a few items. As seen in the following Pareto chart, we analyzed each ergonomically poor level to determine how many items would need to be moved to achieve a substantially positive impact. For both rack level Level 1 and Level 2, a few items represent most potential improvement for the entire level.

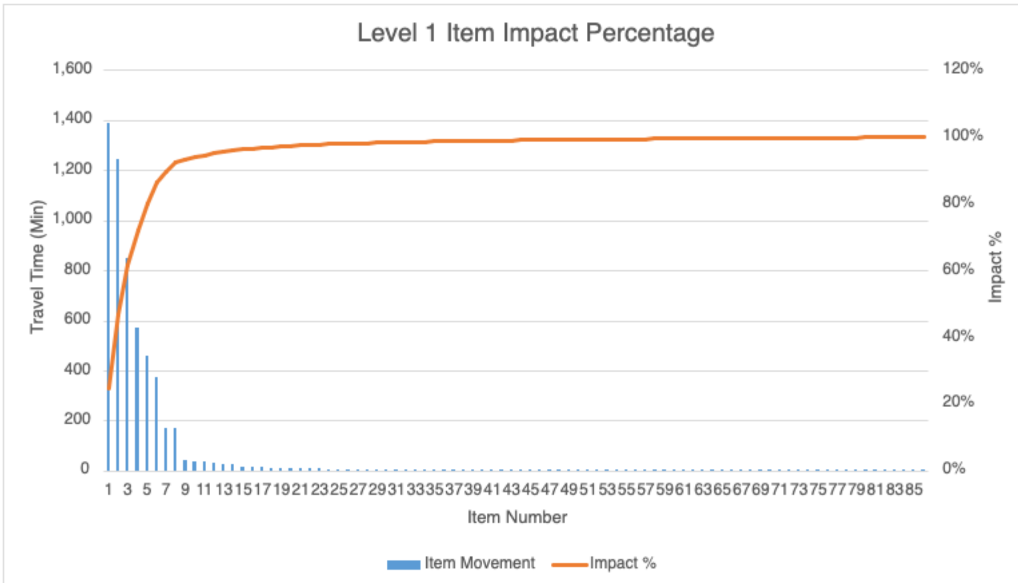

Figure 1: Level 1 Item Pareto Chart

Aisle and Bay Congestion

Within the OFX warehouse, items purchased by Publix are stored in a reserve location using a FIFO inventory methodology. As the pharmaceutical companies which provide the products change their product codes and item codes, the distributors will have to update their systems accordingly. Due to the large amount of product Publix keeps on hand, Publix needs to keep a record of prior item codes as both new and old item codes may be shipping at once. Due to this constraint, new items are slotting into new select locations rather than replacing their predecessor. While this function of the system cannot be changed, it effectively means that when a golden zone item has an item code change, if there is not another golden zone locations available for the updated item code, it will be slotted in levels 1-2. Over the course of the past 10 years, this process has led to many items being displaced without any oversight.

Near Dormant Item Occupation

The 104 items shipped in five or fewer weeks represent a specific, identifiable misallocation of rack resources. These items have not been removed from the active pick area despite their demand profiles indicating that they are either discontinued, or subject to highly intermittent demand. If each of those 104 items occupies a rack location that is within the golden zone, that is valuable real estate being lost. The root cause is the absence of a periodic inventory review mechanism that would flag items falling below a minimum activity threshold for removal. In the current system, items remain in their assigned locations until a manual review identifies them for removal, and no evidence in the data suggests that such a review has occurred during the 52-week observation window, since all 104 low-activity items remain in the dataset as active slot occupants.

Summary

Table 7 consolidates the three root causes identified in this section, linking each to its evidence base in the dataset and quantifying its magnitude. This table serves as the analytical foundation for the design alternatives presented in the Improve section.

Performance Gap	Root Cause Category	Evidence from Data	Magnitude
High movement items outside of the golden zone.	Slotting is done without any velocity data, using only available space as the deciding factor.	394 Fast and Medium moving items slotted outside the golden zone.	Fast and Medium movers make up 96.1% of all facility volume.
Slow moving items in the golden zone.	No periodic review process to identify if items are no longer moving as fast as before.	948 Slow items slotted within the golden zone (some can stay due to excess available space).	Slow moving items only make up 3.9% of all facility volume but represent 49.7% of item codes.
104 near-dormant golden zone items	No effective inventory reviewing process.	Items shipped less than 5 of 52 weeks.	3.5% of golden zone spaces potentially wasted.

Table 7: Summary of Root Causes

The three root causes are interconnected rather than independent. The absence of a velocity-driven slotting rule is the primary driver, and the remaining causes represent specific manifestations of the same underlying failure: slotting decisions were made or maintained without a structured, data-driven methodology that accounts for velocity, container type, weight, and activity simultaneously. A comprehensive re-slotting plan that addresses all root causes through a unified optimization framework will deliver compounding improvements across all KPOVs simultaneously, rather than marginal gains from addressing each cause in isolation.

The Analyze phase confirms that the data are sufficient, consistent, and internally valid to support the design of a materially improved slotting arrangement. The analyses in this section provide direct, quantitative justification for each of the preliminary design alternatives discussed in the Improve phase.

DMAIC: Improve

The design alternatives were developed directly from the root causes identified in the Analyze phase, where major performance gaps were quantified. Two of the key gaps included velocity misalignment and congestion imbalance. To better understand item activity distribution and its impact on slotting efficiency, a summary analysis was performed:

Movement	Items	% of Items	% of Volume	Recommended Level
Fast	89	2.9%	32.2%	3-5

Medium	1432	47.4%	63.9%	3-5
Slow	1500	49.7%	3.9%	1-2
Total	**3021**	**100%**	**100%**	

Table 8: Item Movement Classification

As shown in the table above, approximately 3% of items account for over 32% of total volume, while near 50% contribute less than 4%.

This imbalance confirms that most of the opportunities lie within moving a select few highly impactful products. Based on this finding, the following alternative slotting strategies were developed.

Demand Driven Slotting Strategy

This approach assigns items to rack levels based on demand (bottles/week percentiles). Fast moving items are prioritized for the golden zone, which are levels 3-5, with the goal of increasing ergonomic efficiency.

This directly addresses the primary root cause identified in the analysis: the absence of a velocity driven slotting rule, which resulted in 22% of fast movers being placed in non-optimal levels.

Some limitations are that this approach does not consider weight or dimensional constraints, which may result items moving locations that are difficult to replenish.

Capacity Driven Slotting Strategy

This design prioritizes physical feasibility by assigning items based on dimensions, lane capacity, and storage limits. It ensures that all items fit within their assigned locations and maximizes space utilization.

This approach would be better suited for the warehouse at a later stage when capacity is limited. Due to the excess space currently in the section, this would not be the optimal choice.

Integrated Ergonomic Slotting Strategy

This design combines:

- Velocity using percentiles
- Weight constraints
- Physical fit based on dimensions
- Operational rules

Key design rules:

- Fast moving items - Levels 3-5 (golden zone)
- Slow moving items - Levels 1-2
- Oversized items - Level 5 (constraint-based placement)

- Remaining golden zone - medium demand items

This design directly addresses all major root causes:

Root Cause	Solution Mechanism
Fast Items in Poor Location	Reassign to golden zone
Congestion Imbalance	Redistribute across aisle
Box Sequencing Issue	Enforced constraint
Inactive Items	Removed or deprioritized

Table 9: Root Cause Solution Mechanisms

Tradeoff and Feasibility

To evaluate the feasibility and effectiveness of each alternative, a comparative assessment was performed using key engineering and managerial criteria, including cost, safety, maintainability, and feasibility.

The demand-driven and capacity-based approaches are easier and cheaper to implement, but they don't fully address safety or overall efficiency. The integrated ergonomic approach performs better overall by balancing demand, safety, and feasibility, even if it requires more effort to implement.

Recommended Solution

The Integrated Ergonomic Slotting model is recommended as the optimal solution.

This approach is optimal because it:

- Simultaneously optimizes efficiency, safety, and feasibility
- Directly eliminates all identified root causes
- Produces measurable improvements in:
    - Pick time
    - Workload balance
    - Ergonomic risk

Unlike single-factor approaches, this model ensures that improvements are not isolated but system-wide and sustainable. The final design satisfies key system requirements by integrating demand prioritization, ergonomic safety, and physical feasibility into a single, optimized slotting strategy.

Quantitative Comparison

The proposed model was iteratively refined by adjusting slotting rules and validating results against key performance metrics including pick time, level distribution, and workload balance. Using the developed Excel model, the proposed solution was evaluated against baselines performance metrics.

Pick Time Improvements

- Prior: 81.77 hrs/day
- New: 76.19 hrs/day
- Reduction: 6.82%

Level Rebalancing Impact

- Level 5 (golden zone) increased from 30% to 32% bottles picked
- Level 4 (golden zone) increased from 23% to 30% bottles picked
- Level 3 (golden zone) increased from 24% to 31% bottles picked
- Level 2 decreased from 22% to 7% bottles picked
- Level 1 decreased from 1% to 0% bottles picked

These results demonstrate that the proposed design not only improves efficiency but also redistributes workload toward ergonomically optimal zones. This indicates improved ergonomic alignment, shifting workload toward optimal picking zones.

Workload Redistribution

Heatmaps show:

- Reduced concentration in high-density aisles
- More balanced throughput across bays

This directly addresses throughput imbalance identified in the Analyze section. These results demonstrate that the proposed design not only improves efficiency but also redistributes workload toward ergonomically optimal zones.

Labor Impact

- Annual hours saved: **2,031 hours**

Implementation Plan

Step 1) Data Validation

- Confirm item dimensions, weight, and demand metrics
- Validate against WMS and IE team

Step 2) Model Development

- Use Excel-based tool to:
- Calculate percentiles
- Rank items
- Assign optimal levels

Step 3) Validation

- Compare:
    - Pick time
    - Level distribution
    - Congestion heatmaps

Step 4) Phased Re-Slotting

- Start with top percentile items
- Implement gradually to avoid disruption

Step 5) Continuous Evaluation

- Monitor:
    - Pick efficiency
    - Congestion levels
    - Ergonomic compliance
- Re-run model periodically as demand changes

Project Cost Analysis

Role	Salary	Cost/Hour	Loaded Cost/Hour	Margin	Billing Rate/Hour
Industrial Engineer	$90,000	$43.27	$62.74	50%	$94.11
Data Scientist	$100,000	$48.08	$69.71	50%	$104.57
Project Manager	$150,000	$72.12	$104.57	50%	$156.85

Table 10: Team Cost Breakdown

Phase Estimate	IE Hours	Data Science Hours	PM Hours	Total Cost
Data Cleaning/EDA	0	80	0	$8,365.38
AS-IS KPI	160	40	0	$19,240.38
Optimization Algorithm Dev.	0	40	0	$4,182.69
Ergonomics Modeling	160	0	0	$15,057.69
Validation/Testing	160	40	0	$19,240.38
Presentation & Documentation	160	40	40	$20,808.89
Client Meetings	20	5	5	$3,189.30
Total	**660**	**245**	**45**	**$90,084.74**

Table 11: Project Cost Breakdown

The project hours were developed with a bottom-up work breakdown approach aligned with the major phases outlined in the guidelines. Each phase was defined by its technical scope first, then effort was evaluated based on the level of analytical complexity and required expertise. With the team consisting of one data scientist, four industrial engineers, and one project manager (instructor), responsibilities were divided according to specialization. The data scientist was allocated hours primarily for data cleaning, exploratory data analysis, and optimization algorithm development. The Industrial Engineers were assigned most hours in the AS-IS KPI analysis, ergonomic modeling, validation testing, and documentation, as they require labor intensive system evaluation and operational modeling work. The project manager's hours focused on coordination between the client and firm (UCF) to prepare the project for the team. Hours were distributed to reflect estimated workload by phase, anticipating iteration cycles and unforeseen setbacks. This structured approach ensures labor matches the technical rigor and demand of each phase, while defining clear role accountability amongst the team.

To calculate the return on this project for Publix, the following assumptions were made:

1. All 2,031 hours of calculated savings could be utilized to reduce staffing of the selection area through attrition, reassignment, or capacity absorption.
2. Warehouse selectors are estimated to be compensated $23 per hour.
3. The loaded cost of an associate equates to 1.45 times their wage (bringing selector costs to $33.35 per hour).

Given those assumptions, the following is an estimated return on investment for the project:

Metric	Value
Project Cost (One-Time)	$90,084.74
Annual Labor Savings (Recurring)	$46,713.00
Annual Loaded Labor Savings (Recurring)	$67,733.85
Payback Period (yrs.)	1.3
3 Year ROI	126%
5 Year ROI	276%

Table 12: Estimated Labor Savings and ROI

DMAIC: Control

Control Strategy Overview

The Control phase ensures that the improvements achieved through the re-slotting optimization are sustained over time and continue to deliver measurable performance benefits. Following the DMAIC methodology, this phase focuses on maintaining system stability, preventing regression to previous inefficiencies, and enabling continuous improvement as operation conditions change.

The proposed solution introduced a data-driven slotting methodology based on percentile rankings of item demand, with high-moving items prioritized for placement in ergonomic “golden zone” locations. While the implementation of this solution resulted in improvements in pick time and workload distributions, long-term success depends on the establishment of monitoring systems and a standardized process.

This section outlines the control plan developed to ensure sustainability of the solution, including performance monitoring systems, operational standardization, feedback mechanisms, and risk management strategies.

Process Monitoring and Performance Tracking

To ensure that the improvements achieved through optimization are maintained, a comprehensive performance monitoring system should be implemented. This system will rely on data collected through the warehouse management system (WMS) to continuously track key performance indicators (KPIs) related to picking efficiency and system balance.

The primary KPIs used for monitoring include:

- Total pick time per day
- Average pick time per selector
- Aisle congestion levels (throughput per aisle)
- Rack level utilization and balance
- Percentage of high-frequency items in the golden zone

These metrics should be visualized through dashboards or heatmaps that allow the Industrial Engineering (IE) team and operations supervisors to monitor performance in real time. Deviations from target performance levels should be flagged, which will enable timely corrective action.

To validate the effectiveness of the proposed re-slotting strategy, the optimization was applied using a percentile-based approach. Items above the 60th percentile for movement were relocated to the golden zone and swapped with lower percentile items. This targeted approach ensured that high-frequency items were placed in ergonomically optimal positions while maintaining system balance.

The results of this optimization demonstrated measurable improvements:

- 504 Item Moves
- Total Pick Time/Day reduced by 5.58 Hours
- Overall Pick Time reduced by 6.82%

These results confirm that the solution provides meaningful operational improvements while remaining feasible within existing system constraints. Additionally, these outcomes establish a performance baseline that can be used for ongoing monitoring in the Control phase.

Periodic performance reviews should be conducted on a weekly and monthly basis to compare actual performance against baseline and target values. If performance begins to decline, re-slotting adjustments or further analysis should be conducted to restore system efficiency.

Standardization and Operational Controls

To ensure long-term sustainability of the solution, standardized operating procedures (SOPs) will be developed and implemented. These procedures will define the rules and guidelines for maintaining the optimized slotting configuration and ensuring consistency across the system.

Key standardization elements include:

- Placement of high-frequency items in the golden zone (levels 3-5)
- Assignment of low-frequency items to non-ergonomic zones (levels 1-2)

- Proper container sequencing (bottle before box)
- Maintaining balanced workload distribution across aisles

These guidelines will serve as the foundation for all future slotting decisions and will be documented in formal SOPs accessible to both the IE team and operations staff.

Training programs should be implemented to ensure that all relevant stakeholders understand the new slotting methodology and their roles in maintaining it. This includes training for:

- Industrial Engineering team members responsible for analysis and updates
- Operations supervisors responsible for implementation and enforcement
- Warehouse associates to ensure awareness of system changes

In addition, a structured change management process will be established to handle system updates. This is particularly important for addressing:

- Introduction of new SKUs
- Changes in demand patterns
- Seasonal fluctuation

All changes to the slotting planogram should follow a standardized review and approval process to ensure consistency and prevent unintended disruptions to system performance.

Accountability Structure

Clear accountability is essential to ensure that the control plan is consistently executed and maintained over time. Responsibility for sustaining the optimized slotting system will be shared between the Industrial Engineering (IE) team and operations management.

The Industrial Engineering team will be responsible for monitoring system performance, analyzing KPI trends, and updating the slotting planogram as needed. This includes conducting periodic reviews of demand data, evaluating system balance, and initiating re-slotting adjustments when performance deviates from target levels.

Operations supervisors will be responsible for ensuring adherence to slotting guidelines on the warehouse floor. This includes verifying that items are placed in correct locations, communicating updated to associates, and monitoring day-to-day execution of the slotting strategy.

By establishing a clear ownership, it will ensure that the system does not degrade over time, and that corrective actions are taken promptly when issues arise.

Feedback and Continuous Improvement Systems

To support long-term sustainability, feedback loops will be implemented to enable continuous improvement of the slotting system. These feedback mechanisms allow the solution to adapt to real-world conditions and evolving operational needs.

Feedback should be collected through multiple channels, including regular performance review meetings, KPI monitoring dashboards, and direct input from warehouse associates and supervisors. Operators can provide insight into congestion, accessibility, and picking challenges that may not be fully captured through data alone.

This feedback should be communicated to the Industrial Engineering team, who will evaluate the information and determine whether adjustments to the slotting planogram are necessary. This creates a continuous improvement cycle where the system is regularly evaluated and refined based on both quantitative data and real-world experience.

By incorporating structured feedback loops, the solution becomes dynamic, ensuring that performance improvements are sustained and enhanced over time. This continuous improvement aligns with Industrial Engineering principles by ensuring the system remains adaptable, data-driven, and responsive to operational changes.

Risk Management

Risk management is an important part of this project to ensure that potential issues are identified early and properly addressed. Since this project focuses on optimizing the slotting plan to reduce pick time and congestions, there are several uncertainties that could affect the accuracy of the analysis and the feasibility of the final recommendations.

		Consequence				
		Negligible 1	Minor 2	Moderate 3	Significant 4	Severe 5
Likelihood	Very Unlikely 1	Low	Low	Low	Medium	Medium
	Unlikely 2	Low	Medium	Medium	High	High
	Possible 3	Low	Medium	High	High	Extreme
	Likely 4	Medium	High	High	Extreme	Extreme
	Very Likely 5	Medium	High	Extreme	Extreme	Extreme

Figure 3: Risk Management Matrix

To evaluate the project risks in a structured and consistent way, a risk assessment matrix was created. Each identified risk was evaluated on two factors: likelihood and consequence.

$$Risk\ Score = L \times C$$

This risk matrix helps the team focus on the most critical risks and develop appropriate mitigation plans to ensure the project remains on schedule and achieves its objective of improving picking efficiency. By continuously monitoring and addressing risks throughout the project, the team can increase the likelihood of successfully delivering and effective and implementable slotting optimization.

Risk No.	Risk Description	Likelihood	Consequence	Risk Score	Impact	Actions
R1	KPI data may be inaccurate or incomplete.	2	5	10	High	Verify the data using multiple sources and confirm accuracy.
R2	Limited access to WMS software of necessary data.	3	4	12	High	Coordinate early with Publix and be sure to request all data needed.
R3	Optimization algorithm may not properly improve all KPIs.	3	5	15	High	Test the optimization algorithm/approach on sample data and compare results to current planogram before finalizing.
R4	New slotting plan could create unintended congestion.	2	5	10	High	Evaluate aisle balance and review the planogram carefully before presenting the final solution.
R5	Item demand volume may not remain consistent throughout the year.	3	4	12	High	Historical demand data will be analyzed to discover any seasonal trends or fluctuations. Solution will be designed to be easily updated if demand shifts.
R6	New medications are added last minute and are not included in the algorithm.	4	4	16	Extreme	Algorithm/solution will be designed to be reused and updated when new items are introduced. By incorporating flexible slotting locations, new items can have available locations.
R7	ABC classification may oversimplify item importance.	3	4	12	High	Combine ABC classification with additional KPIs such as pick frequency and congestion.

R8	Unexpected changes to project scope or requirements.	2	4	8	High	As changes from the client are communicated, project schedule will be updated. Included extra time for delays.
R9	Project time constraints may limit full analysis.	3	3	9	High	As a team, we will plan, set deadlines, and prioritize important work.
R10	Unbalanced aisle workload after re-slotting.	3	4	12	High	Validate aisle balance metrics after optimization and compare against baseline.
R11	Incorrect item dimension or capacity calculation.	2	4	8	Medium	Validate item dimensions and capacity calculations with real system constraints.
R12	Underutilization of available space after optimization.	3	3	9	Medium	Incorrect utilization as a constraint in the planogram model.
R13	Failure to achieve target pick time reduction (10%) after optimization.	2	5	10	High	Validate improvements using baseline comparison and sensitivity analysis.

Table 13: Risk Assessment Matrix

Reliability Management FMEA

To support long-term resilience with the proposed slotting solution, a Failure Modes and Effects Analysis (FMEA) was used as a part of the control phase. FMEA is a preventative risk assessment tool that identifies possible ways the improved process could fail, evaluates the effect of each failure on the system's performance, and prioritizes corrective actions before the problem becomes severe.

The purpose of applying FMEA was to evaluate the reliability of the proposed re-slotting process and ensure that the improvement remains effective after implementation. Potential failure modes were identified based on the optimization approach, implementation process, and operational constraints observed in the AS-IS analysis. Each failure mode was then evaluated using the standard FMEA criteria of Severity (S), Occurrence (O), and Detection (D) on a 1-10 scale. These values were multiplied to calculate the Risk Priority Number (RPN):

$$RPN = S \times O \times D$$

Potential Failure Mode	Potential Failure Effect	Potential Causes	S	O	D	RPN	Actions Recommended
Demand data becomes outdated.	Slotting no longer reflects true item velocity.	Demand shifts over time without planogram updates	8	6	4	192	Review demand data monthly and rerun ranking analysis.

New SKUs are not added strategically	New items are placed in non-optimal locations	No formal update process for new products	8	7	5	280	Create change management procedure.
High frequency items remain outside the golden zone	Ergonomic strain and pick time increase	Incomplete implementation of limited available slots	7	5	4	140	Audit golden zone compliance after implementation
Aisle workload becomes unbalanced	Congestion shifts from one aisle to another	Optimization improves level placement but not aisle balance	7	5	5	175	Track aisle throughput and adjust after rollout
Staff do not follow updated slotting procedures	System gradually returns to inefficient state	Lack of training or unclear SOPs	7	4	6	168	Implement training and updated SOP documentation
Performance improvements are not sustained	Labor savings decrease over time	No long-term monitoring or ownership structure	8	5	5	200	Assign process owner and monitor KPIs through dashboards

Table 14: FMEA Control Plan

The FMEA shows that the highest-priority risks are those related to new SKU integration, long-term performance sustainability, and outdated demand data, since these issues could significantly reduce the effectiveness of the solution if not addressed. These risks are especially important because the slotting system is not static; it must continue adapting to changes in product and demand patterns. For that reason, the control plan should not rely on one-time implementation. Instead, it should include periodic reviews, formal update of procedures, and clear ownership of the process.

Summary and Conclusions

This project addressed a longstanding performance gap in the Publix OFX Legends selection area: a carton flow rack slotting planogram that had not been revised since the facility opened in 2016, resulting in systematic misalignment between item placement and item movement demand. The Lean Six Sigma DMAIC methodology provided the analytical framework for the project, and the full-year WMS dataset covering 3,021 slotted item codes across 5,040 rack locations provided the empirical foundation for all findings and improvement recommendations.

The Measure phase confirmed the primary performance gap. Under the current arrangement, 22.7% of all pick volume occurs at Levels 1 and 2, which require repeated floor-level bending and represent the most physically demanding positions in the rack system. Inter-aisle workload was also imbalanced, with aisle LF carrying 26.8% more total pick volume than aisle LB despite comparable physical capacity.

The Analyze phase established that these conditions are not attributable to random variation or data noise. They reflect a systemic absence of a data-driven slotting methodology that periodically realigns item placement with current movement patterns. The primary root cause identified is the failure to reassign item codes to rack positions that reflect their actual demand velocity as that demand has evolved over time. Without a structured review process, the planogram drifted away from optimal alignment as demand changed, resulting in the documented ergonomic and workload imbalances.

The proposed TO-BE slotting redesign addresses these root causes through a targeted repositioning of approximately 16% of item codes within the existing rack infrastructure, requiring no capital expenditure. The methodology applied a composite percentile rank to each item code, averaging individual percentile scores across average bottles per week, average cases per week, and minimum replenishment trips per week. Items ranking in the top 60th percentile of movement currently assigned to Levels 1 or 2 were identified as relocation candidates, and items ranking in the bottom 30th percentile currently occupying golden zone positions were identified as swap targets. The heatmap dashboard developed in the EDA workbook was used to identify specific bay locations with the greatest misalignment and to guide the final swap assignments.

The projected results of the proposed planogram are as follows. The percentage of total pick volume in the golden zone (Levels 3, 4, and 5) increases from 77.3% to over 90%, excluding the LM aisle, representing a 20% improvement in ergonomic pick distribution. The percentage of picks occurring at strenuous Levels 1 and 2 decreases by 66%, from 22.7% to approximately 8%. Overall picking time is reduced by 6.82%, from 81.77 to 76.19 hours per day. The combined improvement is equivalent to a workload reduction of approximately 1 FTE. These gains are achieved by moving approximately 16% of items, representing an estimated 65% of the theoretically optimal solution defined as all non-LM items migrating to the golden zone.

The primary limitation of this work is that performance projections have not yet been validated against post-implementation operational data. Additionally, the solution was developed without access to data on whether items currently occupy multiple facing lanes in the carton flow rack, which introduces uncertainty to the precise extent of improvement achievable. The team scoped the solution deliberately to account for these unknowns, presenting a targeted, implementable re-slot that minimizes operational risk rather than a theoretically complete overhaul requiring data not available to the project team.

Despite these limitations, the analytical case for intervention is strong. The current slotting arrangement demonstrably underperforms relative to the facility's own ergonomic objectives, and the 40.1% rack vacancy provides sufficient flexibility to execute the proposed re-slot entirely within the existing footprint. The team recommends that Publix implement the proposed targeted re-slot, capture pre- and post-implementation pick time data to validate the projected improvements, and establish a periodic review process to update the planogram as demand patterns evolve over time.

Future Work

This project developed a data-driven slotting strategy to improve picking efficiency, reduce congestion, and enhance ergonomics in the Legends section. While the results show measurable improvements, several assumptions and limitations create opportunities for future work.

A key assumption is that item demand remains stable over time. In reality, demand varies due to seasonality and the introduction of new items. Future work should incorporate demand updates, allowing the slotting model to be periodically refreshed using data from the warehouse management system (WMS). This would support continuous optimization and maintain long-term performance.

The current analysis also simplifies operational behavior by not explicitly modeling picker travel paths and congestion interactions. Although workload heatmaps were used, future work should include simulation-based analysis to better capture aisle congestion, picker movement, and system flow under realistic conditions.

Additionally, the current approach relies on rule-based slotting logic. While practical, it may not fully capture complex trade-offs between demand, space constraints, and ergonomics. Future extensions could explore optimization-based methods to further refine slotting decisions.

The risk assessment highlighted important areas for further evaluation, including demand variability, the introduction of new items, and potential congestion after re-slotting. Future work should include scenario testing to validate system performance under different conditions.

A second phase is recommended to support implementation and validation. This phase would require a team of 2–3 Industrial Engineers responsible for validating data, refining the model, coordinating phased implementation, and monitoring key performance indicators such as pick time and workload balance.

Overall, future work should focus on transitioning from a static, analysis-based model to a dynamic, system-integrated solution that continuously adapts to changing operational conditions while maintaining improvements in efficiency, safety, and usability.

Factors Considered

Sustainability is a key factor in our analysis and improvements to the inventory picking system. The requirements of the system are highly likely to change over time based on changing medication demand and growing numbers of stores utilizing the facility. Therefore, the solution we use should be based on substantial historical data as well as focusing on future sustainability of these improvements. While the improvements may fade over time, establishing a proven methodology of easily implemented adjustments to the system may be easily molded to meet changing demand. In a broader sense, our reshuffling of the system will allow for further growth for Publix Pharmacy. The redesigned system will also substantially reduce labor requirements, reducing cost for a vitally important industry.

Safety considerations were most important to our project, with patients and Publix team members as the primary groups potentially impacted by safety concerns. If items were relocated in the system but were unable to be physically moved their new location, products may be improperly placed leading to incorrect picks. This could result in medication shortages. To combat this, we limited movement to entire bays to ensure all item moves would fit in their new locations.

Ethics Analysis

Ethical concerns for workers are generally aligned with economic incentives for this system, since more ergonomic selection points are also faster to pick from. The modifications made to the system aim to improve worker health through reduced picking from ergonomically poor locations while also yielding time savings for the company, which will likely reduce the number of new hires they need to make. Finally, more efficiently using the space allocated to this process will delay the need for a new or expanded warehouse, reducing the environmental impact of this service. The team was also careful to mitigate any potential damage from changes to the system by limiting changes to only strictly beneficial ones. Rather than potentially scrambling item locations, the modular bay-level approach the team took to reorganization will translate easily into the real-world implementation.

While global factors play a limited role in the system itself since Publix is contained entirely in the United States, other applications of the system design should consider this. For a system in China, where Pharmacy volume is likely much higher, more emphasis should fall on ensuring the system can grow with increasing demand. Cultural factors are often overlooked when making engineering decisions. The team focused on using our differing backgrounds to our advantage, each of us bringing a different perspective to problems. At Publix, the company's generally worker friendly attitude may conflict with the demanding nature of the work within this system. Therefore, the company makes up for this with competitive pay and allowing for associates to choose their pace of work.

We considered social factors carefully during our redesign of the system, as they pose a barrier to implementation of a solution. First and foremost, a design created by students will be met with skepticism by any management team. We focused on easy to receive changes for everyone involved. For picking staff, they will notice improved ergonomics at their job, management will observe notable labor savings from reduced picking time, and the operations team will be more willing to implement changes which require minimal investment.

Since the system being designed is not a production system, and does not involve motorized movement of goods, the team does not anticipate substantial environmental impacts from the redesigned system.

Economic consideration was the primary focuses of the project, as we aimed for substantial gains with minimal tradeoffs. To achieve this goal, we largely kept the same system structure while limiting item movement as much as possible. This allows for a minimum amount of investment, likely taking less than one shift of overtime for a few associates to see the expected gains. We generally chose a more conservative approach which limited upfront expenses at the cost of more potential gains.

Sources of Knowledge

Each member of the team used their work and school experiences to shape the methodologies used throughout the project. However, some elements of the UCF curriculum stood out for this project specifically:

Operations Research I

Operations Research I was most impactful for our optimization of the inventory system, as it lays the foundations of linear programming and the key concepts of maximizing improvement while minimal tradeoffs. This project was a unique application of a classic assignment problem, where we effectively treated shelving locations as workers who could perform a single pick at a certain speed. We then found the most misplaced products and reassigned the minimal number to make the most substantial impact. The practical implementation of linear programming problems we learned enabled us to make difficult decisions between optimal placement and upfront investment cost.

Systems Simulation

Systems simulation not only introduces simulation software but also encourages students to think of the physical world as a series of simple tasks. Rather than many different steps within a process, an employee simply moves through a series of quantifiable activities which should be optimized as much as possible. This methodology enabled the team to visually represent the varying inventory of medications as a simple number on a heat map. By equating all items to their most important features, the team was able to understand the real-world impact of changes to the system while also enabling specific methodologies to be implemented.

Quality Engineering

When designing important systems such as pharmacy order assembly stations, quality of the result is incredibly important. Process mapping enabled the team to understand the primary areas for improvement as well as other concerns which might present quality or productivity risks.

Quality Engineering also taught the team Lean Six Sigma tools which were applied to ensure that any changes would not impact the accuracy of the system. Since the system is already functioning to a high degree of accuracy, we implemented several measures to ensure that changes would not cause new problems by limiting variation between comparable elements of the system. This allowed the team to swap elements of the system without causing any new opportunities for error.

Standards and Best Practices

When redesigning a system as important as the warehouse supplying all Publix pharmacies, it was important to the team to ensure alignment with best practices and regulations. To accomplish this, the team used several common Standards and Best Practices to evaluate both the improvement process and the final proposed solution.

Material Handling Institute – Fundamentals

The Material Handling Institute has extensive documentation on best practices and principles for material handling systems such as this picking system. The team used these principles as a starting point for what elements we should focus on. These included ergonomics, standardization, space utilization, and limiting work. While these ideas are general, they should be followed as guiding ideas behind the project, with each action taken by the team focusing on one of these goals.

ISO 11228-1 - Ergonomics Best Practices

The team utilized the principles of ISO 11228-1, which establishes guidelines for safe lifting and lowering practices to prevent worker injury. These guidelines outline the need to reduce lifting from below knee height, which aligns with our goal to reduce volumes from lower shelves due to higher picking times for those levels. The team also analyzed this standard closely during some hard decisions related to the highest level. Although the highest level has some ergonomic concerns from excessive reaching, many items were limited to the top rack since they exceed the height of other racks. Therefore, many compromises were already required within the system. Since the team lead identified a negligible pick time difference between Level 5 and the golden zone, the team was able to treat level 5 as effectively equivalent to the golden zone. Using the ergonomics principle that working against gravity is more challenging than working with it, the team relocated more frequently picked items from low levels to Level 5. While no picking system can be perfect, this choice will reduce the likelihood of injury for warehouse employees.

Standard warehouse Design Practices

These standards emphasize balanced workload distribution to reduce congestion. The analysis revealed significant imbalance across aisles, with certain areas experiencing concentrated activity while others remain underutilized. This indicates a need for improved distribution of high-demand items. If high-demand items are not spread across aisles, workers will stack up in certain regions, adding queue time to their process flow which is entirely waste. Spreading these high demand items reduces the likelihood that two workers need to access the same shelving area at the same time. Efficient use of available space is a key design standard. With 59.9% utilization, the system has sufficient capacity for current demand, suggesting that performance issues are caused by poor item distribution rather than space limitations. This supports an approach focused on optimization within the existing layout rather than expansion.

Acknowledgements

Our team would like to thank all individuals who contributed their time, expertise, and guidance to the success of this project.

We extend our appreciation to our faculty mentors, Dr. Luis Rabelo and Dr. Mansooreh Mollaghasemi, for their continued instruction, mentorship, and support throughout this project. Their guidance in professional presentation development, project planning, and problem-solving was instrumental in strengthening both the technical quality of our work and our overall professional growth.

We would especially like to thank Ernesto Martinez De La Pena, Manager of Industrial Engineering at Publix, for serving as our primary point of contact and providing continuous support. His industry knowledge, project direction, and insight into Publix’s operational environment were essential in helping our team understand the system, validate our analysis, and develop practical and implementable solutions.

Finally, we are grateful to the Industrial Engineering team at Publix for providing us with the opportunity to work on this project. This experience allowed us to apply academic knowledge in a real-world setting while further developing our technical, analytical, and professional skills.

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Appendix

SENIOR DESIGN PROJECT CHARTER

Project Sponsor:	*Publix Supermarkets Inc.*		
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Co – Project Manager			
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Process Impacted:	*OFX Legends Selection Section*		
Expected Start Date:	*January 3rd 2026*		
Expected Completion Date:	*April 30th 2026*		
Green Belts Assigned:	*Luis Rabelo, Ph.D.*	Black Belts Assigned:	*Ahmed Mohammed Ph.D.*
Mentor:	*Mansooreh Mollaghasemi, Ph.D.*	Master Black Belt Assigned:	*Jose Flores, Ph.D.*

Element	Description of Element	Project Charter
Title of Project	What is the title of the improvement project?	*OFX Legends Re-Slot Analysis*

Element	Description of Element	Project Charter
Project Description	Brief background and project's purpose	*The Publix Pharmacy Distribution Center (OFX), located in Orlando Florida, serves as the primary distribution hub for pharmaceutical products to more than 1,400 Publix Pharmacy locations across Florida, Gorgia, North Carolina, South Carolina, Alabama Tennessee, Virginia, and Kentucky. In addition to supporting retail pharmacies, OFX also supplies Central Fill operations, making it a critical component of Publix's pharmaceutical supply chain. The facility must maintain a high level of accuracy, speed, and reliability to ensure timely replenishment of essential medications across the network.* *Since the facility opened in 2016, the Legends slotting plan has not been modified. If slotting practices are not periodically evaluated and improved, the OFX Legends selection area will suffer from inefficiencies in congestion and labor scheduling. Given the scale and criticality of this distribution center, improving efficiency presents an opportunity to enhance productivity while maintaining compliance with physical and ergonomic constraints.*
Process Description	Brief description of the process in which opportunity exists	*Within OFX, there are four different selection areas Refrigerated, Vault, Cage, and Legends, each containing different items that are shipped simultaneously. The Vault, and Cage are controlled areas that store medications with higher regulations. The area we are interested in is Legends, this area contains 3,832 unique item codes distributed across 5,040 flow rack locations. Selectors pick items manually from the rack locations to build store orders. When the item is picked, it is placed in a tote which the selector carries using a cart that is manually pushed.* *The legends area consists of aisles that are composed of racks, which are ordered alphabetically starting from LA to LM. Each aisle has enough space so two selectors can fit side by side with their carts. Due to high volume of daily picks, slotting decisions directly influence selectors' travel time, rack-level accessibility, aisle congestion, and ergonomic conditions. Even small inefficiencies in item placement can accumulate across thousands of picks, affecting overall labor.* *The slotting plan must also operate within defined operational constraints. These include rack-level weight limits, case height clearances, ergonomic placement considerations for fast-moving items, balanced movement across rack bays to reduce congestion, and prioritization of plastic bottles over boxes to prevent damage. These constraints increase the complexity of slotting decisions and require a structured, data-driven approach.*

<table>
<tr><th>Element</th><th>Description of Element</th><th colspan="3">Project Charter</th></tr>
<tr><td>Business Case</td><td>Brief explanation of the expected financial improvement, or other justification</td><td colspan="3">Addressing these issues in the current slotting layout through a data-driven optimization approach offers clear operational benefits. By reorganizing high-movement items and balancing workload across aisles, the proposed solution is expected to reduce average pick time by roughly 10% and decrease congestion. These improvements directly support Publix's goals for accuracy and efficiency in pharmaceutical distribution. From an ergonomic perspective, placing fast-moving items in the golden zone and following weight and height limits helps reduce bending, reaching and overall physical strain on employees. The resulting slotting framework is also reusable, allowing Publix to update the layout as demand patterns change.</td></tr>
<tr><td>Benefit to External Customers</td><td>Who are the final customers and what benefits will they see?</td><td colspan="3">The final customers will be the OFX operations team. We expect them to see benefits in labor hours spent picking, as well as congestion within the aisles.</td></tr>
<tr><td>Problem Statement</td><td>Problem and goal statement (project's purpose)</td><td colspan="3">The goal of this project is to improve order selection efficiency withing the Legends section in the Publix Pharmacy Distribution Center by developing and evaluating a data-driven slotting plan that reduces average pick time and balances selector workload. By reducing ergonomic and physical constraints, the team can:

Balance stocker workload.

Reduce aisle congestion, measured by the item movement per rack bay.

Reduce the average pick time per unit by at least 10% by the end of the project period.

Each objective is measurable using performance metrics derived from item movement data and rack level pick time parameters. These performance targets represent preliminary improvement goals and will be validated and adjusted as necessary.</td></tr>
<tr><td rowspan="5">Project Objectives</td><td rowspan="5">What are the performance metrics such as throughput, quality, inventory, defects, yield, costs, etc.? What improvement is targeted?</td><td>Performance
Metric</td><td>Current
(As-Is) Value</td><td>Future
(To-Be) Value</td></tr>
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Element	Description of Element	Project Charter
Scope Statement	Which parts of the process will be investigated and which parts will be excluded? What are the major tasks to complete?	*This project is aimed at reducing the time spent picking items from a flowrack selector section. The goal is to slot the items such that there is minimal congestion between selectors in the aisles, while also reducing the number of times products will be picked from inconvenient locations. The following section lays out what will be in/out of scope for the project. Key tasks and assumptions will also be listed.* *In Scope:* *Analyzing the current planogram for the flowrack by visiting the site and meeting with IE and operations teams.* *Calculating baseline KPIs based on the original planogram, which will be used to measure improvements.* *Presenting the findings of the AS-IS analysis, setting expectations for potential improvements given the current state of the system.* *Optimizing the slotting plan to achieve improvements across all KPIs.* *Presenting the optimized plan, including both labor and cost savings for the new planogram.* *Out of Scope:* *Changing any part of the selection process or Publix operations.* *Changing any part of the facility racking/flowrack layout.* *Making recommendations on labor reductions due to time saved.*

Proposed Methods	Which design and analysis concepts, techniques and tools are going to be used?	Item Movement Analysis & Classification *Historical item movement data will be analyzed to quantify pick frequency and volume for each SKU. Items will be classified using ABC analysis, a standard inventory and operations management techniques.* *A items: highest movement and pick frequency* *B items: moderate movement* *C items: low movement* *This classification enables prioritization of placement decisions based on operational impact rather than subjective judgment. ABC analysis is appropriate because it directly links demand behavior to layout efficiency and labor productivity.* *Slotting & Layout Design Methodology* *Slotting decisions will be made using Industrial Engineering layout and ergonomics principles.* *"A" items will be placed in the ergonomic golden zone to minimize bending, reaching, and walking.* *The highest moving items will be removed from carton flow racks and assigned to pallet pick locations to reduce congestion and handling time.* *Lower rack levels will be reserved for lighter and slower moving items.* *Plastic bottles will be slotted before boxed items to prevent box damage during picking.* *This methodology improves pick speed, reduces worker fatigue, and minimizes product damage while respecting physical constraints such as rack clearance and mezzanine load limits.* *Ergonomic & Weight Constraints* *Ergonomic principles are applied to reduce musculoskeletal risk by limiting excessive reaching, bending, and repetitive motion. Weight limits of 14 lbs. per square foot per carton flow level are enforced to maintain mezzanine structural safety. Case height constraints are incorporated to ensure compatibility with rack clearances and prevent unsafe handling conditions.* *These considerations directly reduce injury risk, lost time incidents, and noncompliance with safety regulations.* *Congestion Reduction & Workload Balancing* *High velocity items will be deliberately distributed across multiple aisles and rack bays rather than clustered. This reduces picker interference and aisle congestion. Row level balancing will be used to distribute stocking and picking workload evenly, preventing bottlenecks during peak periods.*

Element	Description of Element	Project Charter
		Picker routes will be evaluated and adjusted to reduce unnecessary travel distance and cross aisle traffic. These actions apply IE principles of flow optimization and workload leveling. *Data Validation & Performance Measurement* *The proposed layout will be validated using historical warehouse data and scenario-based analysis.* *Key performance metrics include:* *Average pick time* *Picker travel distance* *Congestion frequency* *Stocker workload distribution* *Comparisons between the current and proposed states will be used to quantify expected improvements before operational implementation. This reduces technical and operational risk.*
Key Deliverables	What are the tangible and/or intangible objects that are produced as a result of the project and that relate directly to and align with the objectives of the project?	*Excel Sheet with Updated Racking Order* *Update provided list of items and rack locations with optimized value for easy implementation by customer* *Heat Map of Picking Activity* *Heat map based on current racking layout* *Updated Heat Map with new racking positions* *KPI Projections Data Sheet* *This Data sheet will include valuable data to predict improvements to the system including:* *Expected balance of racks* *Change in number of items picked from each shelf level* *The balance of Stocker shelves* *Projected improvements to picking speed based on changes* *Final Presentation* *Current-state findings* *Methodological explanation* *Overview of expected benefits to customer including heat maps*

Element	Description of Element	Project Charter	
Schedule	Give the key project milestones and dates	Project Start Date:	*January 23rd 2026*
	"D" – Define Phase	"D" Completion Date:	*2/23/26*
	"M" – Measure Phase	"M" Completion Date:	*3/13/26*
	"A" – Analysis Phase	"A" Completion Date:	*3/13/26*
	"I" – Improve Phase	"I" Completion Date:	*4/17/26*
	"C" – Control Phase	"C" Completion Date:	*Ongoing*
Team Members	Names of the project team members (in alphabetical order of last names)?	*Nathan James, Cristina Morales, Ivan Rey Pestana, Diego Rubio, Misael Castellanos Valido*	
Costs and Resources	What are the estimated project costs and needed resources?	*No cost to Publix or UCF, Publix IE team provided dataset for analysis.*	